Asymptotically Harmonic Partitions of the Harmonic Series

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Abstract

We arrange the positive integers in a triangular array and sum reciprocals down columns. Each column gives a convergent reciprocal series with strictly convex denominators. If $S^{(k)}$ denotes the kth column sum, then the harmonic series decomposes as

$$\sum_{n=1}^{\infty} \frac{1}{n} = \sum_{k=1}^{\infty} S^{(k)},$$

while the column sums themselves satisfy $S^{(k)} > 2/k$ for every k and in fact $S^{(k)} \sim 2/k$. This yields a self-similar proof that the harmonic series diverges. This note records a triangular-column variant of the harmonic-series divergence proof, offering a structurally distinct perspective on the classical literature.

1 Introduction

The harmonic series

$$\sum_{n=1}^{\infty} \frac{1}{n}$$

has accumulated a remarkably varied literature of elementary divergence proofs, valued as much for viewpoint and pedagogy as for technical depth. Kifowit and Stamps [1] surveyed many such arguments, and Kifowit's later note [2] enlarged that collection further to 45 distinct proofs. Among the recurring motifs are Oresme-type blockings, Bernoulli-style contradiction arguments, triangular-number groupings, telescoping comparisons, and very short proof variants such as those of Bradley [3], Cusumano [5, 6], and Baker [4]. Historical and expository discussions, such as Dunham's account of the Bernoullis [7], further illustrate the richness of this classical topic. That survey is hard to improve upon as a catalog of entertaining arguments.

The present note adds another member to this expository family: a triangular-array argument obtained by reading the usual row-wise decomposition by columns instead. The resulting column subseries have strictly convex denominators and convergent sums, but the sequence of their total masses is itself asymptotically harmonic, yielding a self-similar contradiction proof of divergence.

The point is not to compete with the classical proofs in simplicity, but to exhibit a different structural feature of the harmonic series: a layered partition into convergent convex subseries whose total masses remain asymptotically harmonic.

2 The triangular array

Let

$$\mathrm{T}_m := \frac{m(m+1)}{2}, \qquad m \geq 0,$$

with the convention $\mathrm{T}_0 = 0$. Arrange the positive integers in rows of lengths $1, 2, 3, \ldots$:

$$\begin{array}{ccccc} 1 & & & & \\ 2 & 3 & & & \\ 4 & 5 & 6 & & \\ 7 & 8 & 9 & 10 & \\ 11 & 12 & 13 & 14 & 15 \\ \vdots & & & & \end{array}$$

Thus row m consists of the integers

$$\mathrm{T}_{m-1} + 1,\ \mathrm{T}_{m-1} + 2,\ \ldots,\ \mathrm{T}_m,$$

for $m \geq 1$. For each fixed $k \in \mathbb{N}$, let $S^{(k)}$ be the sum of reciprocals down the kth column. Since the kth column begins in row k, we have

$$S^{(k)} := \sum_{m=k}^{\infty} \frac{1}{\mathrm{T}_{m-1} + k}. \tag{1}$$

The first few examples are

$$\begin{aligned} S^{(1)} &= 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{7} + \frac{1}{11} + \cdots, \\ S^{(2)} &= \frac{1}{3} + \frac{1}{5} + \frac{1}{8} + \frac{1}{12} + \frac{1}{17} + \cdots, \\ S^{(3)} &= \frac{1}{6} + \frac{1}{9} + \frac{1}{13} + \frac{1}{18} + \cdots. \end{aligned}$$

Since every positive integer appears exactly once in the triangular array, these column series partition the harmonic series.

Proposition 1. *One has*

$$\sum_{n=1}^{\infty} \frac{1}{n} = \sum_{k=1}^{\infty} S^{(k)}.$$

Proof. The sets of denominators appearing in the columns are pairwise disjoint and their union is $\mathbb{N}$. Since all terms are positive, the identity follows simply by regrouping terms according to their column index. □

3 Convex denominator sequences

The first column is

$$1, 2, 4, 7, 11, 16, \ldots,$$

which is the minimal strictly convex increasing sequence of natural numbers, namely

$$a_n = 1 + \frac{n(n-1)}{2}.$$

More generally, every column has strictly convex denominators.

Proposition 2. *Fix $k \geq 1$. The denominators in $S^{(k)}$ are*

$$d_m^{(k)} := \mathrm{T}_{m-1} + k \qquad (m \geq k),$$

and satisfy

$$d_{m+1}^{(k)} - d_m^{(k)} = m, \qquad d_{m+2}^{(k)} - 2d_{m+1}^{(k)} + d_m^{(k)} = 1.$$

In particular, each column is a convergent reciprocal series with strictly convex denominators.

Proof. Since $\mathrm{T}_m - \mathrm{T}_{m-1} = m$, the first identity is immediate:

$$d_{m+1}^{(k)} - d_m^{(k)} = (\mathrm{T}_m + k) - (\mathrm{T}_{m-1} + k) = m.$$

Taking another difference gives the constant second difference 1. For convergence, note that $d_m^{(k)} \geq \mathrm{T}_{m-1} \sim m^2/2$, so

$$\sum_{m=k}^{\infty} \frac{1}{d_m^{(k)}}$$

converges by comparison with a constant multiple of $\sum m^{-2}$. □

A convenient closed form for the denominators is

$$\mathrm{T}_{m-1} + k = \frac{m^2 - m + 2k}{2},$$

so (1) may also be written as

$$S^{(k)} = \sum_{m=k}^{\infty} \frac{2}{m^2 - m + 2k}. \tag{2}$$

After the index shift $n = m - 1$, this becomes

$$S^{(k)} = \sum_{n=k-1}^{\infty} \frac{2}{n^2 + n + 2k}. \tag{3}$$

4 A lower bound for the column sums

The key estimate is pleasantly simple.

Theorem 1. *For every $k \geq 1$,*

$$S^{(k)} > \frac{2}{k}.$$

Proof. Starting from (3), compare the nth term with a telescoping one. For $n \geq k - 1$,

$$(n+1)(n+2) - (n^2 + n + 2k) = 2(n + 1 - k) \geq 0.$$

Hence

$$\frac{2}{n^2 + n + 2k} \geq \frac{2}{(n+1)(n+2)} = 2\left(\frac{1}{n+1} - \frac{1}{n+2}\right).$$

Summing from $n = k - 1$ to ∞ gives

$$S^{(k)} \geq 2 \sum_{n=k-1}^{\infty} \left(\frac{1}{n+1} - \frac{1}{n+2}\right) = \frac{2}{k}.$$

The inequality is strict because equality holds only for the first term $n = k - 1$; for all $n \geq k$ one has $2(n + 1 - k) > 0$. □

5 A self-similar proof of divergence

The previous theorem immediately gives a contradiction proof.

Theorem 2. *The harmonic series diverges.*

Proof. Assume for contradiction that

$$\sum_{n=1}^{\infty} \frac{1}{n} = M$$

for some finite M. By Proposition 1,

$$M = \sum_{k=1}^{\infty} S^{(k)}.$$

By Theorem 1,

$$\sum_{k=1}^{\infty} S^{(k)} > \sum_{k=1}^{\infty} \frac{2}{k}.$$

Under the assumption that the harmonic series converges to M, the right-hand side equals $2M$. Therefore

$$M > 2M,$$

a contradiction. □

A compelling feature of this proof is that the harmonic series is decomposed into convergent pieces, but the sequence of their total masses is still large enough to dominate a constant multiple of the harmonic series itself.

6 Asymptotics

In fact, the lower bound of Theorem 1 is asymptotically sharp.

Proposition 3. *As $k \to \infty$,*

$$S^{(k)} \sim \frac{2}{k}.$$

Proof. The lower bound $S^{(k)} > 2/k$ is already known. For an upper bound, split off the first term in (3) and then compare the remaining denominator with $n(n+1)$:

$$S^{(k)} = \frac{2}{k(k+1)} + \sum_{n=k}^{\infty} \frac{2}{n^2+n+2k} < \frac{2}{k(k+1)} + \sum_{n=k}^{\infty} \frac{2}{n(n+1)}.$$

Since

$$\sum_{n=k}^{\infty} \frac{2}{n(n+1)} = \frac{2}{k},$$

we obtain

$$\frac{2}{k} < S^{(k)} < \frac{2}{k} + \frac{2}{k(k+1)}$$

The error term is $O(k^{-2})$, so $S^{(k)} = \frac{2}{k} + O(k^{-2})$, and in particular $S^{(k)} \sim \frac{2}{k}$. □

Thus the harmonic series admits a decomposition into convergent convex subseries whose total masses are themselves asymptotically harmonic.

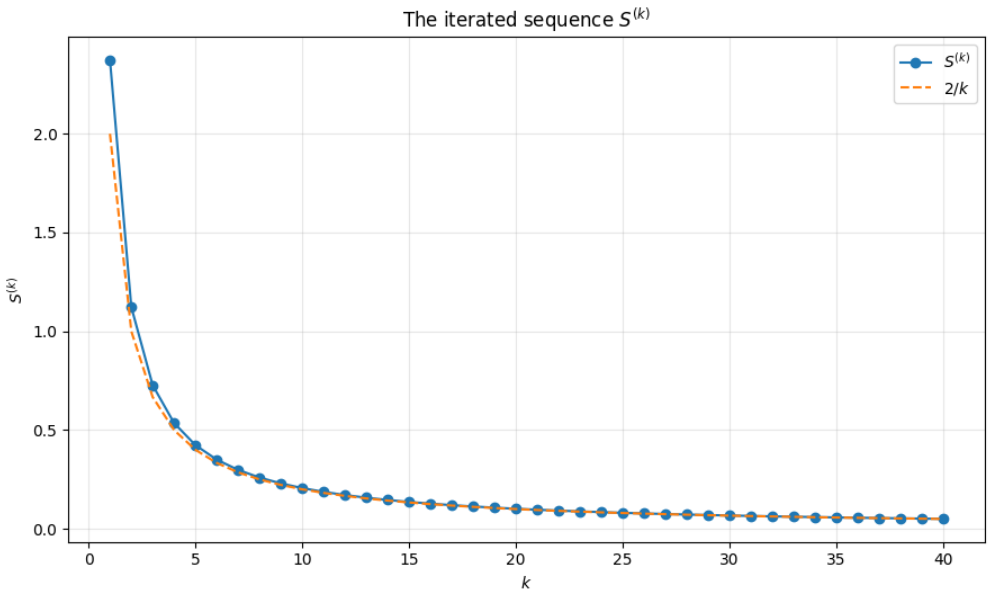

Figure 1: The column sums $S^{(k)}$ plotted against k, with the asymptotic curve $2/k$ overlaid. The sequence drops quickly and then tracks $2/k$ closely, consistent with the asymptotic relation $S^{(k)} \sim 2/k$. The first few values are $S^{(1)} \approx 2.37365$, $S^{(2)} \approx 1.12229$, $S^{(3)} \approx 0.72680$, and $S^{(4)} \approx 0.53564$.

7 A few remarks and variants

Remark 1. *The first column sum is*

$$S^{(1)} = 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{7} + \frac{1}{11} + \cdots = \sum_{n=1}^{\infty} \frac{1}{1 + \frac{n(n-1)}{2}}.$$

A short calculation with partial fractions and the digamma function yields the pleasant closed form

$$S^{(1)} = \frac{2\pi}{\sqrt{7}} \tanh\left(\frac{\pi\sqrt{7}}{2}\right) \approx 2.3736546754.$$

So the largest column sum is already finite and quite small.

Remark 2 (Closed forms for the column sums)**.** *More generally, for every $k \geq 1$,*

$$S^{(k)} = \sum_{m=k-1}^{\infty} \frac{2}{m^2 + m + 2k}.$$

By separating off the first $k-1$ terms from the full series, one obtains for $k \geq 2$

$$S^{(k)} = \frac{2\pi}{\sqrt{8k-1}} \tanh\left(\frac{\pi\sqrt{8k-1}}{2}\right) - 2\sum_{m=0}^{k-2} \frac{1}{m^2 + m + 2k}.$$

Equivalently, writing ψ for the digamma function and $\Im$ for the imaginary part,

$$S^{(k)} = \frac{4}{\sqrt{8k-1}} \Im\psi\left(k - \frac{1}{2} + \frac{i\sqrt{8k-1}}{2}\right).$$

Thus each column sum can be written explicitly, and the first few values are

$$S^{(1)} \approx 2.37365, \qquad S^{(2)} \approx 1.12229, \qquad S^{(3)} \approx 0.72680, \qquad S^{(4)} \approx 0.53564.$$

While these closed forms are not needed for the divergence argument, they reinforce the sense that the triangular decomposition has a surprisingly rigid and explicit structure.

Derivation of the closed forms. Let

$$\Delta_k := \sqrt{8k-1}.$$

Then

$$m^2 + m + 2k = \left(m + \frac{1}{2} - \frac{i\Delta_k}{2}\right)\left(m + \frac{1}{2} + \frac{i\Delta_k}{2}\right),$$

so

$$\frac{2}{m^2+m+2k} = \frac{2}{i\Delta_k}\left(\frac{1}{m+\frac{1}{2}-\frac{i\Delta_k}{2}} - \frac{1}{m+\frac{1}{2}+\frac{i\Delta_k}{2}}\right).$$

Using the digamma recurrence

$$\psi(z+1) - \psi(z) = \frac{1}{z},$$

see [9, (5.5.2)], we obtain for $N \geq k-1$,

$$\sum_{m=k-1}^{N} \frac{2}{m^2+m+2k} = \frac{2}{i\Delta_k}\Big[\psi\Big(N+\frac{3}{2}-\frac{i\Delta_k}{2}\Big) - \psi\Big(N+\frac{3}{2}+\frac{i\Delta_k}{2}\Big) - \psi\Big(k-\frac{1}{2}-\frac{i\Delta_k}{2}\Big) + \psi\Big(k-\frac{1}{2}+\frac{i\Delta_k}{2}\Big)\Big].$$

As $N \to \infty$, the first two terms cancel by the standard asymptotic expansion of ψ, see [9, (5.11.2)]. Hence

$$S^{(k)} = \frac{2}{i\Delta_k}\left[\psi\Big(k-\frac{1}{2}+\frac{i\Delta_k}{2}\Big) - \psi\Big(k-\frac{1}{2}-\frac{i\Delta_k}{2}\Big)\right] = \frac{4}{\Delta_k}\Im\psi\Big(k-\frac{1}{2}+\frac{i\Delta_k}{2}\Big).$$

Now let

$$F_k := \sum_{m=0}^{\infty} \frac{2}{m^2+m+2k}.$$

The same computation with lower limit 0 gives

$$F_k = \frac{4}{\Delta_k}\Im\psi\Big(\frac{1}{2}+\frac{i\Delta_k}{2}\Big).$$

Using the identity

$$\Im\psi\Big(\frac{1}{2}+iy\Big) = \frac{\pi}{2}\tanh(\pi y),$$

see [9, (5.4.17)], we obtain

$$F_k = \frac{2\pi}{\Delta_k}\tanh\Big(\frac{\pi\Delta_k}{2}\Big).$$

Since

$$S^{(k)} = F_k - 2\sum_{m=0}^{k-2} \frac{1}{m^2+m+2k} \qquad (k \geq 2),$$

this yields

$$S^{(k)} = \frac{2\pi}{\sqrt{8k-1}}\tanh\Big(\frac{\pi\sqrt{8k-1}}{2}\Big) - 2\sum_{m=0}^{k-2}\frac{1}{m^2+m+2k}.$$

For $k=1$, the finite correction term is absent. □

Remark 3. *There is a close family resemblance between the present argument and several proofs discussed by Kifowit and Stamps [1]. In particular:*

- *their Proof 6 is the Gillman–Honsberger pairing argument, with the observation that analogous contradictions arise by grouping in threes, fours, and so on; see also Chowdhury [8] and the discussion in [2];*
- *their Proof 13 is the Johann Bernoulli telescoping argument, which derives a contradiction of the form $S = 1 + S$; see also Dunham [7];*
- *their Proof 14 groups the harmonic series by triangular-number rows and obtains the contradiction $S > 2(S - 1)$.*

While the present note shares a family resemblance with these grouping arguments, its vertical, column-wise partition offers a structurally distinct perspective.

8 Conclusion

The triangular array of positive integers carries two natural directions. Reading by rows gives one of the classical triangular-number proofs of the divergence of the harmonic series. Reading by columns gives a different picture: one obtains convergent reciprocal series with strictly convex denominators, but their total masses remain harmonically large. The resulting proof highlights an elegant and largely unexplored fractal-like property of the harmonic series.

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